

EpiCenter/LIS Minimal Standard Interface Requirements

The BD "LIS Vendor Interface Document" (L-005933) describes all of the data fields and data exchange capabilities that EpiCenter & BD Instruments can handle through an LIS Interface. While it may be difficult, or undesirable for an LIS Vendor to comply with all of the specifications set forth in that document, there are certain capabilities that are inherent and necessary in implementing a useful EpiCenter/LIS interface. These are described below in Section 1.

Note: This document is intended as a companion to the "LIS Vendor Interface Document".

1. Features of the Standard EpiCenter LIS Interface (Minimal Requirements)

This section will describe the features necessary to implement a "Standard" EpiCenter/LIS interface. These are the minimum capabilities for such an interface.

1.1. Host Query

The LIS shall respond to a query from EpiCenter, containing a single accession number, by downloading data relating to that accession number. Those data fields are described in the table in Section 3.

1.2. Unsolicited Download Demographics

The LIS will not only respond to Host queries with all related demographic data, but will be able to support unsolicited download of demographic data to EpiCenter. It is also desirable for the LIS to be able to transmit new demographic data to EpiCenter as additional or changed information becomes available at the LIS.

1.3. Download Orders

The LIS shall be able to download ordered tests to EpiCenter by including Accession Number, Test ID, Isolate number (for Phoenix and MGIT 960 AST test orders), and organism ID (For Phoenix and MGIT 960 AST tests). It is not necessary to transmit the BD Sequence number in a downloaded test order.

1.4. Test Results

The LIS shall accept and process test result data uploaded from EpiCenter. It shall accept all Result Fields described in the "Required Upload Fields" section below.

The LIS shall be capable of supporting a variable number of drug results from a Phoenix test, including drugs that are not on the panel, which may be added to the test results by the BDXpert system. These results will appear on the isolate results record. (See Uploading Isolate Results)

The LIS shall support rebuilding the AST result set on every transfer from EpiCenter. (Rapid reporting of drug results can/will result in drugs being sent incrementally as the results are produced by the instrument.)

1.5. Mixed Instrumented Test Results

The LIS shall be able to exchange test results from a variety of instruments with EpiCenter at the same time. This means that the test orders and results could be associated with a BACTEC 9000 Growth test, a BACTEC 960 Growth test or AST test, or a Phoenix ID, AST, or ID/AST test .

1.6. Uploading Isolate Results

The LIS shall be able to recognize and process the uploading of isolate results in addition to test results. Isolate results are a combined view of all ID and AST results run through EpiCenter on the same Accession/Isolate. Isolate level results have been “Expertized” by the EpiCenter and provide the most accurate, microbiologically correct results.

Isolate results contain the original test result, the “Expertized” result, and the final result selected by the user (if required by conflict).

The LIS shall handle Resistance Markers as an Isolate Level Result Type (not an organism or a drug)

1.7. Data Fields

1.7.1. Required Download Fields:

These fields represent the minimum required data to be downloaded to EpiCenter to define patient, specimen and/or ordered test.

See the LIS Interface table below in Section 3 for a description of which download fields are required to satisfy the requirements of the Standard LIS Interface. For a further description of these fields, their range of values and their record position, see document L-005933.

1.7.2. Required Upload Fields:

These fields shall be processed by the LIS when uploaded from EpiCenter. At the very least, they can be ignored, but this will limit the data with which the LIS has to work. There is a wide range of data that Epicenter will upload to the LIS, and it is in the best interest of both systems to maximize the processing of this uploaded data. See the LIS Interface table below in Section 3 for a description of which upload fields are required to satisfy the requirements of the Standard LIS Interface. For a further description of these fields, their range of values and their position, see document L-005933.

2. Additional (Optional) Components of the Full Function EpiCenter LIS Interface

This section describes additional capabilities that, while not absolutely essential to EpiCenter/LIS data exchange, would greatly enhance both systems' access to mutually shared data.

2.1. Comments

The LIS shall handle Comment text for both downloads and uploads. Comment types can be Patient, Specimen, Test, Isolate and Expert. This is particularly important with Expert comments that contain important and relevant information about the test and isolate results. The Comment records sent by EpiCenter to the LIS include chartable clinical notes relevant to the organism identified as well as clarifications on why the BDxpert system modified a result.

2.2. Download offline test results

The LIS shall download to EpiCenter offline test results on demand and unsolicited.

2.3. Upload offline test results

The LIS shall accept offline test orders from EpiCenter.

2.4. Data Fields

2.4.1. Optional Download fields

Any or all of these download fields would enhance data exchange between EpiCenter and LIS. These represent remaining fields described in document L-005933 not specified above. See the LIS Interface table in Section 3 below for a description of which download fields are available to enhance the interface. For a further description of these fields, their range of values and their position, see document L-005933.

2.4.2. Optional Upload fields

Any or all of the following upload fields would enhance data exchange between EpiCenter and LIS. See the LIS Interface table in Section 3 below for a description of which upload fields are available to enhance the interface. For a further description of these fields, their range of values and their position, see document L-005933.

3. EpiCenter/LIS Interface Feature/Field Table

This table reflects which features and fields are supported for both upload and download by the Standard and Enhanced EpiCenter LIS interface. NA indicates not applicable. This has been used in such cases as Demographic Uploads where EpiCenter has the capability to upload demographic data, but it is irrelevant to EpiCenter as to whether the LIS processes this data.

Features/Fields	Standard (Minimal) Interface		Enhanced (Full Feature) Interface	
	Upload	Download	Upload	Download
Host Query	Yes	NA	Yes	Yes
Demographics	NA	Yes	NA	Yes
Test Orders	NA	Yes	NA	Yes
Test Results	Yes	No	Yes	NA
Mixed Instrumented Test Results	Yes	No	Yes	NA
Isolate Results	Yes*	No	Yes	NA
Comments	No	No	Yes	Yes
Offline Test Results	No	No	Yes	Yes
Patient Record Fields				
Patient ID	NA	Yes	NA	Yes
Patient Name	NA	Yes	NA	Yes
Date of Birth	NA	Yes	NA	Yes
Patient Sex	NA	Yes	NA	Yes
Diagnosis Code	NA	Yes	NA	Yes
Patient Current Therapy	NA	Yes	NA	Yes
Hospital Service Code	NA	Yes	NA	Yes
Hospital Client	NA	Yes	NA	Yes
Admitting Physician Code	NA	No	NA	Yes
User Defined Patient Field 1	NA	No	NA	Yes
User Defined Patient Field 2	NA	No	NA	Yes
User Defined Patient Field 3	NA	No	NA	Yes
User Defined Patient Field 4	NA	No	NA	Yes
User Defined Patient Field 5	NA	No	NA	Yes
Admission Date and Time	NA	No	NA	Yes
Patient Room Number	NA	No	NA	Yes
Patient Phone Number	NA	No	NA	Yes
Order Record Fields				
Accession Number	NA	Yes	NA	Yes
Isolate Number	NA	Yes	NA	Yes
Organism ID	NA	Yes***	NA	Yes***
Test ID code	NA	Yes	NA	Yes
Collection Date and Time	NA	Yes	NA	Yes

	Standard (Minimal) Interface		Enhanced (Full Feature) Interface	
Features/Fields	Upload	Download	Upload	Download
“Collected By” Code	NA	Yes	NA	Yes
Receipt Date and Time	NA	Yes	NA	Yes
Specimen Type	NA	Yes	NA	Yes
Body Site	NA	Yes	NA	Yes
Ordering Physician Code	NA	Yes	NA	Yes
Test Sequence Number	NA	No	NA	Yes
User Defined Specimen Field 1	NA	No	NA	Yes
User Defined Specimen Field 2	NA	No	NA	Yes
User Defined Specimen Field 3	NA	No	NA	Yes
User Defined Specimen Field 4	NA	No	NA	Yes
User Defined Specimen Field 5	NA	No	NA	Yes
“Received By” Code	NA	No	NA	Yes
Specimen Action Code	NA	No	NA	Yes
Isolate Source Test	NA	No	NA	Yes
Isolate Source Test Start Time	NA	No	NA	Yes
Ordering Physician Phone Numbers	NA	No	NA	Yes
Specimen Reimbursement Value	NA	No	NA	Yes
Test Reimbursement Value	NA	No	NA	Yes
Isolate Classification/Nosocomial	NA	No	NA	Yes
4. Result Record Fields				
Result ID Code	Yes**	NA	Yes	NA
Antibiotic	Yes	NA	Yes	NA
Antibiotic Concentration	Yes	NA	Yes	NA
Units	Yes	NA	Yes	NA
Test Status	Yes	NA	Yes	NA
Test Sequence Number	Yes	NA	Yes	Yes
Result Data Field 1	Yes	NA	Yes	NA
Result Data Field 2	Yes	NA	Yes	NA
Result Data Field 3	Yes	NA	Yes	NA
Result Data Field 4	Yes	NA	Yes	NA
Result Data Field 5	Yes	NA	Yes	NA
Preliminary/Final Status	Yes	NA	Yes	NA
Result Record Fields				
Test Start Date and Time	Yes	NA	Yes	NA
Result Date and Time	Yes	NA	Yes	NA
Test Complete	Yes	NA	Yes	NA
Date and Time	Yes	NA	Yes	NA
Instrument Type	Yes	NA	Yes	NA
Media Type	Yes	NA	Yes	NA
Protocol Length	Yes	No	Yes	Yes
Instrument Number	Yes	NA	Yes	NA

	Standard (Minimal) Interface		Enhanced (Full Feature) Interface	
Features/Fields	Upload	Download	Upload	Download
Instrument Location	Yes	NA	Yes	NA
Request Comment Fields				
Comment Type	No	No	Yes	Yes
Comment Text	No	No	Yes	Yes
Request Record Fields				
Starting Access Number	Yes	Yes	Yes	Yes
Starting Patient ID	No	No	No	Yes
Starting Sequence Number	No	No	No	Yes
Ending Patient ID	No	No	No	Yes
Ending Accession Number	No	No	No	Yes
Ending Sequence Number	No	No	No	Yes
Test ID	No	No	No	Yes
Test Status	No	No	No	Yes
Instrument Type	No	No	No	Yes
Instrument Number	No	No	No	Yes
Result Qualifier	No	No	No	Yes
Time Qualifier	No	No	No	Yes
Starting Date/Time	No	No	No	Yes
Ending Date/Time	No	No	No	Yes

*** Isolate Results Upload:** See Section 1.6 for a description of Isolate Results Uploads

**** Result ID Code Upload:** This field identifies the type of test results are being uploaded in this record. It also determines the subsequent format and content of the data in this record. It must be the first field processed, in order to determine how to process the rest of the data fields in this record. Possible values for this field are GND, GND_MGIT, GND_PROBETEC, AST, AST_MGIT, AST_MIC, ASI_DIA, ID, and OTHER.

*****Organism ID Download:** In the Standard interface, this field shall be downloaded initially for all isolate based tests (i.e. AST). Additionally, in the advanced interface, this field shall also be downloaded subsequently for all G&D tests, once the Organism ID has been determined.